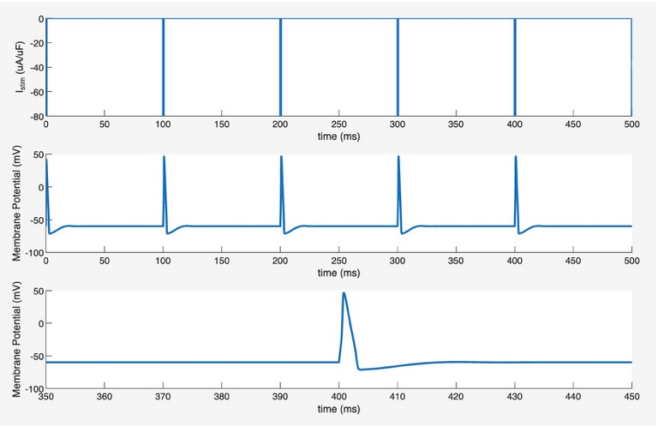
# Computational Medicine — Practical 1

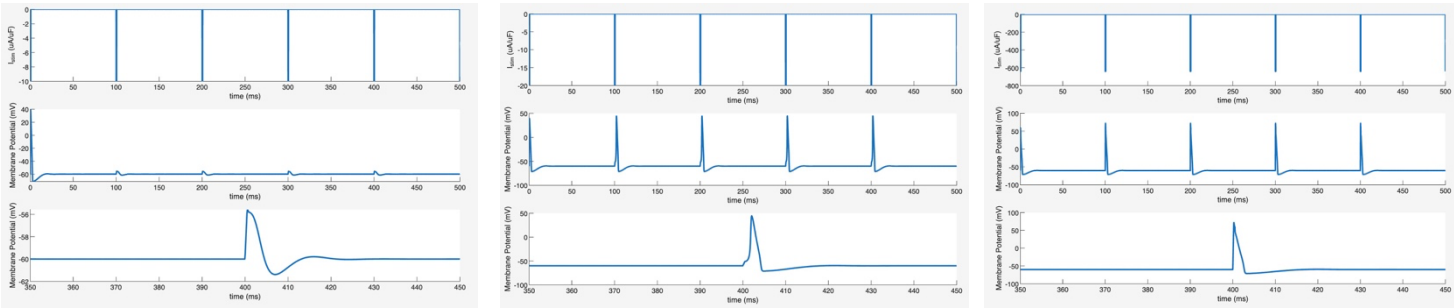
## Section 2 — The Hodgkin and Huxley model

### Stimulus Amplitude

Here are results for varying levels of magnitude for the stimulus current:




Base (80uA)



10uA (left), 20uA (middle), 640uA (right)

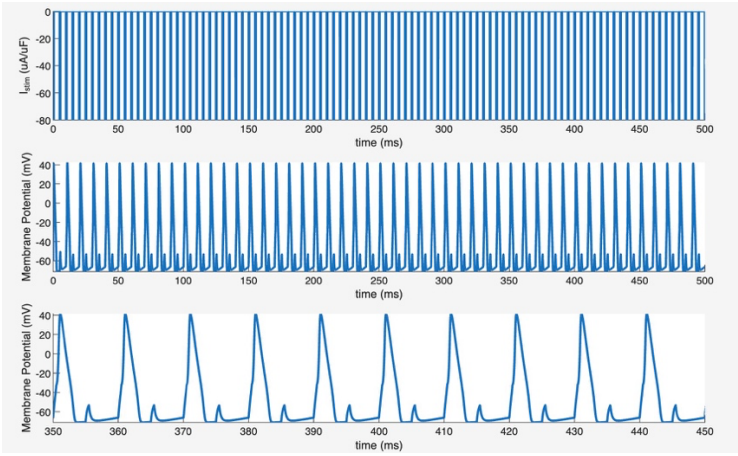
At 80 µA, the membrane potential makes a spikey curve but with only two distinct lines that are sharp and one slow recovery.

When we go too low (≤ 10 µA), the membrane potential is unable to spike properly, staying at around −60 mV after the initial crash instead of oscillating between 40 and −70 mV and replicating the first cycle.

When we increase the intensity (> 500 µA), we can see our stimulus working fine, but the membrane potential decays in 4 steps instead of the 3.

This seems to stay similar with higher intensities.

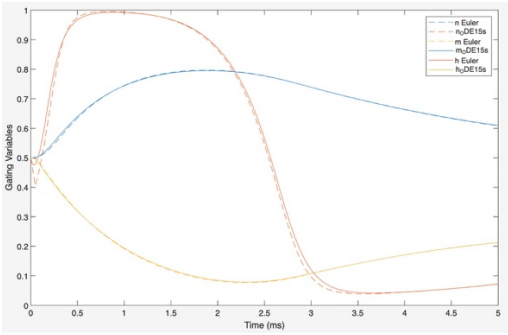
**Stimulus Frequency**




Cycle Length of 5ms

When we decrease the CL to 5 ms and below, another smaller spike appears in between regular spikes.

**Euler's method**




The difference comes from numerical errors from Euler's method, especially in areas where the second derivative is far from 0.

To reduce this difference we can reduce dt or use a more precise method.

<u>Section 4 — Run example script</u>

**State variables**

There are 43 state variables, as shown in the state vector. Each of those values have an associated ODE.

**Ionic currents updating the voltage**

As we can see on line 230, we use:

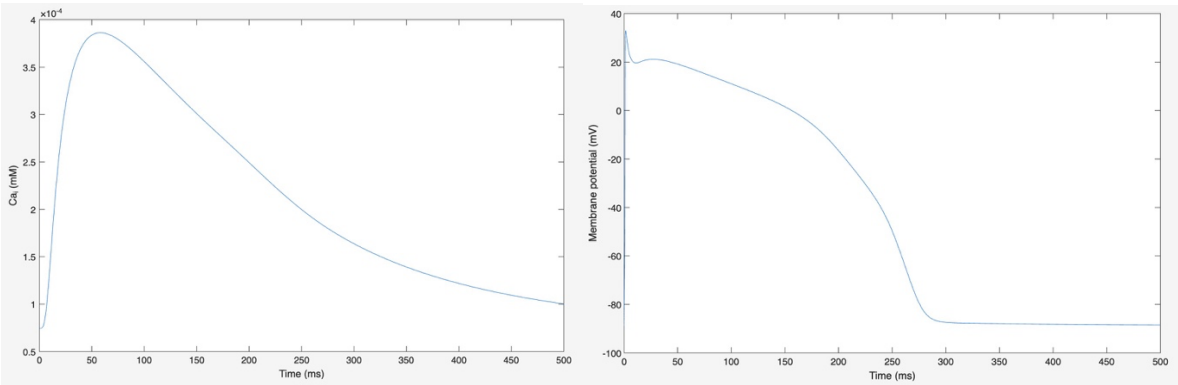
- INa
- INaL
- Ito

- ICaL
- ICaNa
- ICaK
- IKr
- IKs
- IK1
- INaCa_i
- INaCa_ss
- INaK
- INb
- IKb
- IpCa
- ICab
- I_ClCa
- I_Clbk
- Istim

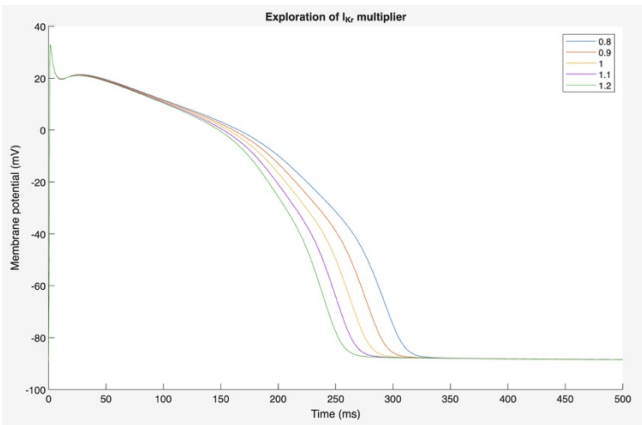
**Ionic current is implemented using a Markov model**

IKr, the intensity of the potassium current (the rapidly activating version) is implemented using a Markov Model

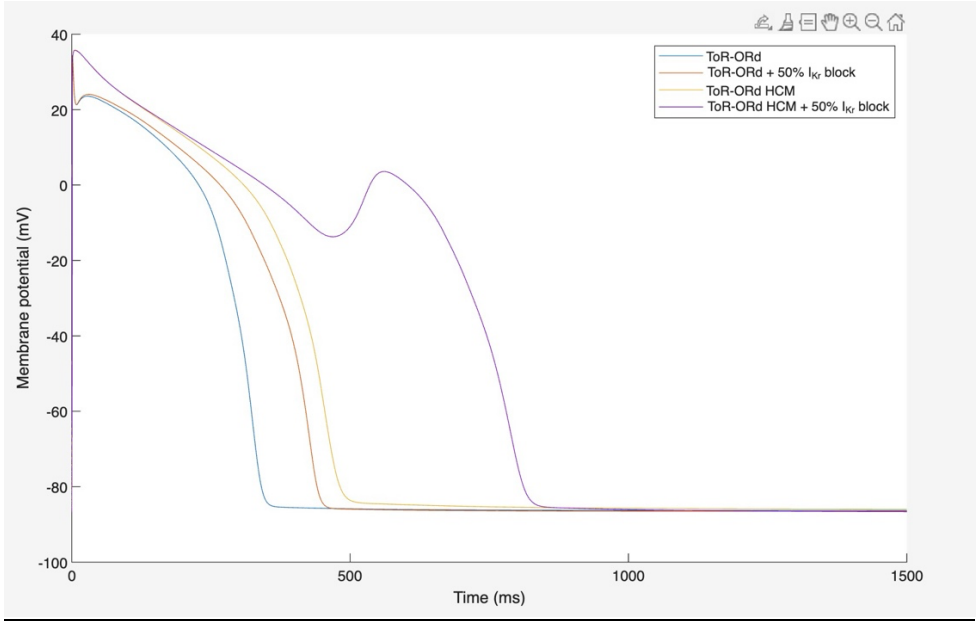
**Script 1 output**




**Script 2 output**

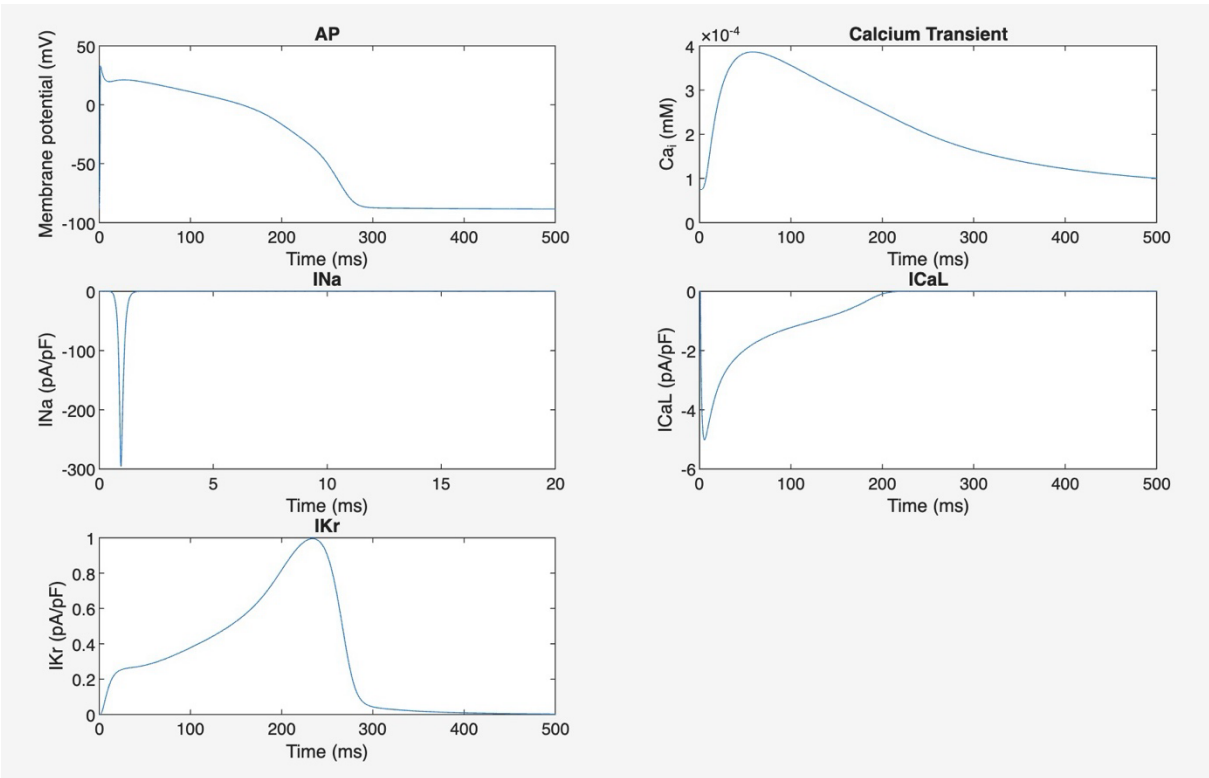


**Script 3 output**




Section 5 — Single-cell simulations

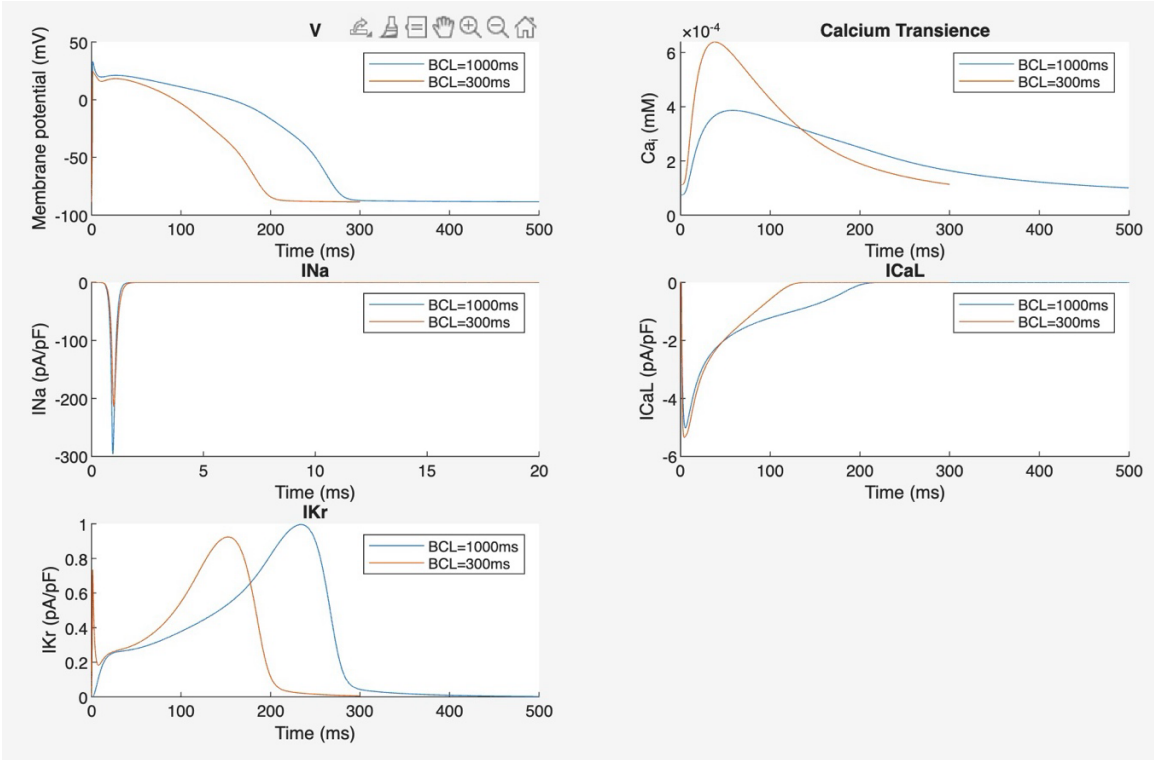
PART A




**Difference between Calcium transient and ICaL current**

The calcium transient is simply the concentration of calcium in the membrane, while the ICaL current shows the opening of the calcium gate of the membrane.

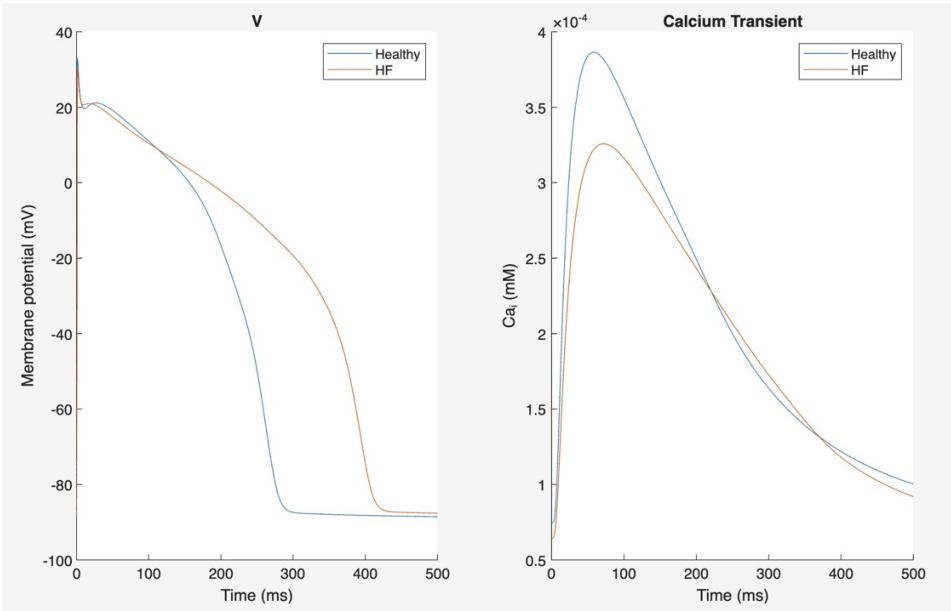
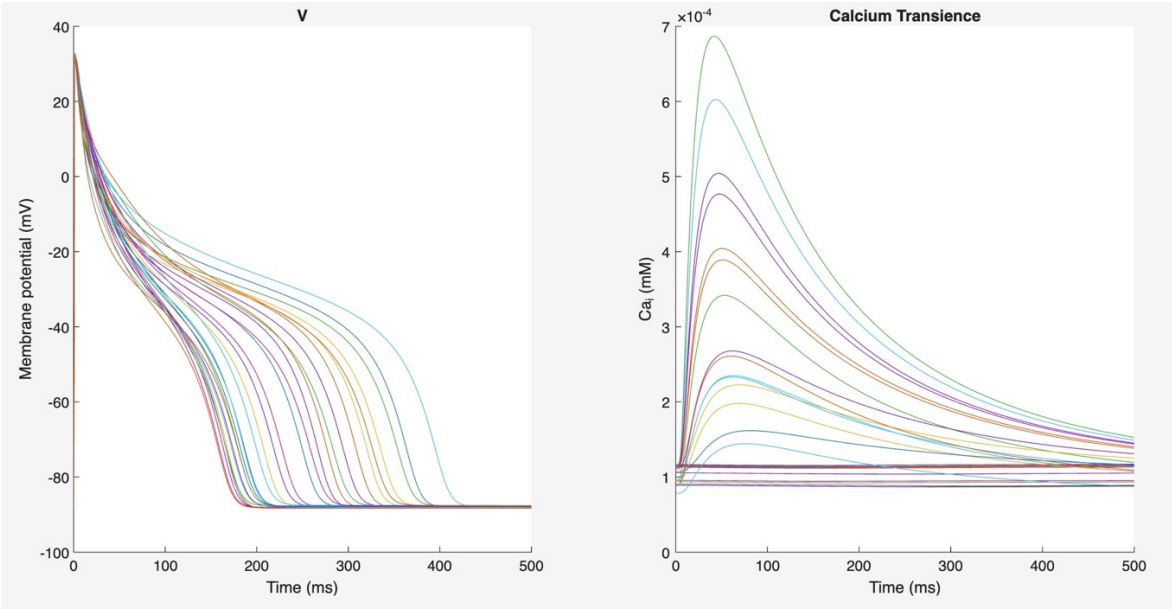
PART B




**Phenomena observed**

- As expected, the voltage and IKr spike earlier.
- The intensity of the Na current doesn't decline as much.
- For Ca_i^{2+}, we have a similar spike with double the amplitude.
- For CaL, we have a faster recovery to 0 pA/pF.

<u>Section 6 — Heart Failure Modelling</u>



Section 7 — Population of Models